

Aptech

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APTECH COMPUTER EDUCATION

HDSE — Semester III | E-Project

BrightWorld LED Lights

Online LED Lighting Store Website

" BRIGHTWORLD — LED LIGHTING PORTAL "

(PREPARED BY)

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Submitted To: Aptech Computer Education

Instructor: Sir Asad Ali

Submission Date: July 27, 2026

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E — Certificate

This is to certify that the following students have successfully completed their **HDSE Semester III E-Project** titled:

" BrightWorld LED Lights — Online LED Lighting Store Website "

MR. HUSSNAIN AHMED, MR. MUHAMMAD ALI, MR. AHMED RAZA, MR. USMAN TARIQ

have demonstrated outstanding dedication, technical skills, and teamwork in developing a fully functional, responsive, and professional web-based LED Lighting Portal.

The project has been reviewed, evaluated, and approved for submission to the **E-Projects Team at Aptech Head Office.**

Submitted to: Aptech Computer Education

Date: July 27, 2026

Student Signature

Instructor: Sir Asad Ali

Center Manager

Acknowledgement

First and foremost, we extend our sincere gratitude to **Almighty Allah** for granting us the strength, patience, and wisdom to successfully complete this E-Project titled *"BrightWorld LED Lights — Online LED Lighting Store Website."* Without His blessings, this achievement would not have been possible.

We would like to express our heartfelt appreciation to our respected instructor, **Sir Asad Ali**, whose continuous support, valuable guidance, and technical mentorship proved to be a pillar of strength for our team throughout the course of this project. Their vast knowledge, insightful feedback, and constant encouragement consistently motivated us to push our limits and explore deeper aspects of web development.

We also gratefully acknowledge the collaborative spirit and relentless dedication of our entire project team:

**MR. HUSSNAIN AHMED | MR. MUHAMMAD ALI | MR. AHMED RAZA | MR. USMAN
TARIQ**

Each member brought unique strengths and actively contributed toward front-end development, design, JavaScript functionality, documentation, and quality assurance. Our team effort, coordination, and mutual support were key to delivering a fully functional, responsive, and visually impressive web-based application.

We would also like to extend our gratitude to our parents and well-wishers for their constant encouragement, emotional support, and understanding throughout the development period. Their motivation helped us stay focused and committed to delivering a high-quality project.

Finally, we sincerely thank the **E-Project Coordination Team at Aptech Head Office** for their consistent communication, clear project structure, and timely support that guided us in the right direction. Their efforts in managing and evaluating such large-scale academic projects are truly commendable.

This project has been a valuable learning experience for us — helping bridge the gap between academic theory and practical application. We welcome constructive feedback and suggestions to improve this project further.

Thank You!

Introduction

The rapid growth of e-commerce and digital retail has fundamentally transformed the way businesses operate and interact with their customers. In today's competitive market environment, even specialized product stores — such as those dealing in LED lighting — require a strong online presence to remain relevant and accessible to a wider audience. BrightWorld, a physical LED lighting retail store, identified this need and sought to develop a comprehensive web-based solution to showcase its products, attract customers, and provide detailed product information online.

The **BrightWorld LED Lights Website** is a fully responsive, multi-page static web application designed to serve as the official online storefront for BrightWorld. This website provides customers with a seamless digital experience, allowing them to browse through various categories of LED lighting products, explore product specifications and pricing, discover different brands, and make informed purchasing decisions — all from the comfort of their homes.

The website features a modern, premium design with a dark-mode interface, smooth animations, an interactive product filtering and search system, a shopping cart, customer feedback functionality, and downloadable product documentation. It is built entirely using front-end technologies — HTML5, CSS3, and JavaScript — with Bootstrap 5 as the CSS framework.

This project not only fulfills the business requirement of establishing BrightWorld's online presence, but also demonstrates the practical application of modern web development skills, including responsive design, dynamic UI interactions, LocalStorage-based state management, and professional-grade visual design.

The website consists of **11 fully developed HTML pages**, each serving a specific purpose — from product listings and brand showcasing to feedback forms, gallery displays, and downloadable resources. Every page is consistent in design, fully responsive across all screen sizes, and follows best practices in web development and user experience design.

What's This E-Project

The **E-Project** is a structured, hands-on learning initiative introduced by Aptech Computer Education that bridges classroom-based theoretical knowledge with real-world practical application. It provides students with a professional platform to design, develop, and document a fully functional software or web application based on a real industry problem.

The E-Project enables students to:

- Practice in a structured and step-by-step manner, ensuring gradual skill development and confidence building.
- Build comprehensive, multi-page web applications that demonstrate proficiency in frontend technologies.
- Utilize various tools, libraries, and frameworks to design visually appealing and functionally efficient applications.
- Transform individual components into a unified, well-organized project with consistent structure and design.
- Learn to implement concepts in phases — requirements, design, development, testing, and documentation.
- Work on real-life scenarios that align with current industry standards and user expectations.
- Tackle realistic business challenges to design meaningful and valuable digital solutions.

For our **BrightWorld LED Lights** project, the E-Project framework provided a strong foundation for developing a functional, creative, and user-friendly website. This phased development approach allowed our team to collaborate effectively while applying modern web development techniques and industry best practices.

Throughout the project, we benefited from continuous mentoring and guidance from our instructor, who supported us in overcoming technical challenges and refining our application to a professional standard. After completion, the project along with its detailed documentation has been submitted to the E-Projects Team at Aptech for evaluation.

This experience has significantly strengthened our technical skills in HTML5, CSS3, JavaScript, and Bootstrap, while also improving our understanding of teamwork, problem-solving, project management, and real-world application development.

Objectives of the Project

The primary objective of this E-Project is to design and develop a fully functional, visually appealing, and user-friendly **Online LED Lighting Store Website** for BrightWorld. This system aims to provide a centralized digital platform where customers can browse LED lighting products, access specifications and pricing, explore brand offerings, submit feedback, and download product documentation — all without visiting the physical store.

More specifically, the objectives of this project are:

- To develop a professional, multi-page website that serves as BrightWorld's complete digital storefront, accessible to customers at any time from any device.
- To display all LED product categories (Bulbs, Spot Lights, Decoration Lights, Smart Lights, Tube Lights, and Ceiling Lights) with images, specifications, and pricing.
- To provide a simple, interactive, and fully responsive user interface that ensures ease of navigation for all types of users on mobile, tablet, and desktop devices.
- To implement a real-time product search, category filter, and sorting system that helps customers quickly find the products they need.
- To showcase multiple LED brands (Philips, Osram, Havells, Syska, Wipro, Bajaj, Orient, GM Modular) with dedicated brand pages and filtered product listings.
- To include a customer feedback form that allows visitors to share ratings, reviews, and suggestions with the store management.
- To provide downloadable product documentation — including Price Lists, Product Catalogues, Technical Specifications, and Warranty Guides — for customer convenience.
- To implement a shopping cart system using LocalStorage that enables customers to add products, manage quantities, and review selections.
- To apply a premium Dark Mode / Light Mode theme toggle with user preference persistence across sessions.
- To apply modern web development tools and best practices learned during the course, including HTML5 semantics, CSS3 animations, JavaScript ES6+, Bootstrap 5, and AOS library.
- To enhance teamwork, communication, and real-world problem-solving skills through collaborative project development within the team.

Problem Statement

Original Client Requirement (as provided):

BrightWorld is having a store for LED Lights. They have a collection of lights. Now with the changing environment, they want their presence on the Internet. Hence they need to have a website with all their products listed on the website. Consumers should find it easy to view various products and decide upon purchasing the right product to meet their requirements. A website needs to be designed to display various lights available with the store. Details of locations, feedback must be included.

In the traditional retail model, customers of BrightWorld LED Lights are required to physically visit the store to browse products, check specifications, compare prices, and make purchasing decisions. This process is often time-consuming and inconvenient, especially for customers living far from the store, those with busy schedules, or individuals who prefer to research products before making a purchase.

The key problems addressed by this project are as follows:

- Customers have no way to browse BrightWorld's complete product inventory online, leading to missed sales opportunities and reduced customer reach.
- There is no digital platform where customers can compare product specifications, pricing, and brands without visiting the physical store.
- The store has no mechanism for collecting customer feedback digitally, making it difficult to improve services based on customer opinions.
- Product documentation such as price lists, technical specifications, and catalogues are not accessible to customers outside the store.
- Customers cannot view the store's location, contact information, or operating hours without directly calling or visiting the store.
- There is no organized way to showcase the multiple LED brands offered by the store, making it difficult for brand-conscious customers to identify their preferred products.
- The lack of an online presence results in reduced visibility in an increasingly digital marketplace, where competitors with websites gain a significant advantage.

Due to these challenges, there is a strong need for a professional, user-friendly, and comprehensive website for BrightWorld. Such a platform will provide a centralized digital storefront where customers can browse products, view specifications and pricing, explore brand offerings, find store contact details, submit feedback, and download product resources — thereby improving customer experience and expanding the store's market reach.

Requirement Specification

The website has been designed and developed based on the following requirement specifications provided by the client. All 10 requirements have been fully implemented in the final build:

#	Requirement	Implementation	Status
1	The Home Page should be created making use of Sections with a suitable logo, the header section can be added which will show images of various LED Lights.	index.html — Full-screen Hero Carousel (4 slides), BrightWorld logo with lightbulb icon, category sections with LED images.	✓ Done
2	The site must contain the links to navigate through various types of LED Lights.	Sticky Navbar with 3 dropdown menus: Shop, Company, Support. Mobile offcanvas slide-in menu.	✓ Done
3	Links of LED Lights can be categorized like (Bulbs, Spot Lights, Decoration Lights, Smart Lights etc.)	products.html — Filter buttons for 6 categories: Bulbs, Spot Lights, Decoration, Smart Lights, Tube Lights, Ceiling Lights.	✓ Done
4	Specification and Pricing of Lights should be added along with the images.	Each product card shows: image, brand, name, spec tags, price in PKR, and Add-to-Cart button.	✓ Done
5	Website should provide section for various brands offered in the store.	brands.html — Dedicated page for 8 brands: Philips, Osram, Havells, Syska, Wipro, Bajaj, Orient, GM Modular.	✓ Done
6	Gallery section should be added for viewing different images.	gallery.html — Filterable image gallery with hover zoom effects and lightbox viewer.	✓ Done
7	Feedback must be allowed to enter by the viewer.	feedback.html — Complete form: name, email, phone, star rating (1–5), category, message.	✓ Done
8	Site map, Gallery, About us, Queries link, Contact us link must be added.	All 5 pages created: sitemap.html, gallery.html, about.html, queries.html, contact.html — linked in navbar	✓ Done

#	Requirement	Implementation	Status
		and footer.	
9	The complete description must be allowed to get download as pdf or doc file.	download.html — 4 documents: Price List, Product Catalogue, Technical Specs, Warranty Guide.	✓ Done
10	The advertise section with various facilities provided should be mentioned.	index.html — Features section, animated stats counters (10K+ Customers, 500+ Products), promotional banners.	✓ Done

Result: All 10 out of 10 requirements have been successfully implemented.

Pages Description

1. Home Page — index.html

The main entry point of the website. Contains: Loading Screen, Top Bar with contact info, Sticky Navbar with dropdowns, Hero Carousel (4 animated slides), Animated Stats Counter (10K+ Customers, 500+ Products, 50+ Brands, 15+ Years), Product Categories section, Featured Products, Brands section, Advertisement Banner, Testimonials, and Footer.

2. Products Page — products.html

Full product listing with: real-time Search Bar, Category Filter Buttons (All / Bulbs / Spot Lights / Decoration / Smart Lights / Tube Lights / Ceiling), Sort Options (Price & Name), Product Cards with images/specs/price, Add-to-Cart functionality, and URL deep-link filtering support.

3. Brands Page — brands.html

Dedicated showcase for 8 brand partners: **Philips (Netherlands), Osram (Germany), Havells, Syska, Wipro, Bajaj, Orient, GM Modular (India)**. Each brand card links to filtered product listings.

4. Gallery Page — gallery.html

Responsive image gallery with filter tabs (All, Interior, Exterior, Commercial, Home Decor), hover zoom/overlay effects, and lightbox for full-size image viewing.

5. About Us — about.html

Company history, mission & vision, team introduction, achievements, and reasons to choose BrightWorld.

6. Feedback Page — feedback.html

Customer feedback form: Name, Email, Phone, Product Category, Star Rating (1–5), Comments field, Submit button.

7. Contact Us — contact.html

Physical store address, map section, phone numbers, email, store hours (Mon–Sat: 9 AM – 10 PM), and direct contact form.

8. Queries & FAQ — queries.html

Accordion-style frequently asked questions and a query submission form for customer support.

9. Sitemap — sitemap.html

Visual HTML sitemap showing complete website page hierarchy for easy navigation reference.

10. Downloads — [download.html](#)

Four downloadable documents: Price List, Product Catalogue, Technical Specifications, and Warranty Guide.

11. Cart — [cart.html](#)

Shopping cart with item display, quantity management, price calculation, and LocalStorage-based persistence.

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Technology Stack

Category	Technology / Tool	Version / Details
Markup Language	HTML5	Semantic elements, SEO meta tags, ARIA labels
Stylesheet	CSS3	Custom Properties, Glassmorphism, Animations, Grid & Flexbox
Scripting Language	JavaScript (ES6+)	Vanilla JS, Strict Mode, LocalStorage API, IntersectionObserver
CSS Framework	Bootstrap	Version 5.3.3 (loaded via CDN)
Icon Library	Bootstrap Icons	Version 1.11.3 (loaded via CDN)
Scroll Animations	AOS — Animate On Scroll	Version 2.3.4 (loaded via CDN)
Typography	Google Fonts	Inter · Poppins · Montserrat
State Management	LocalStorage (Browser API)	Cart items, Theme preference persistence
Development Tool	Visual Studio Code	Code editor with Live Server extension
Browser Testing	Google Chrome	Chrome DevTools for responsive testing

Project Statistics

File	Size	Lines of Code
index.html (Home Page)	67 KB	1,235 lines
products.html	46 KB	672 lines
css/style.css	39 KB	1,400+ lines
js/script.js	16 KB	454 lines
Total HTML Pages	—	11 pages
Total Project Size	~19 MB (with images)	—

File Structure

```
BrightWorld/
├─ index.html      ← Home Page
├─ products.html   ← Products Listing
├─ brands.html     ← Brands Showcase
├─ gallery.html    ← Image Gallery
├─ about.html      ← About Us
├─ feedback.html   ← Feedback Form
├─ contact.html    ← Contact & Location
├─ queries.html    ← FAQs & Queries
├─ sitemap.html    ← Site Map
├─ download.html   ← Downloads
├─ cart.html       ← Shopping Cart
├─ css/
│   └─ style.css   ← Main Stylesheet (1400+ lines)
├─ js/
│   └─ script.js   ← Main JavaScript (454 lines)
├─ assets/img/
│   ├── BULBS/
│   ├── SPOT LIGHTS/
│   ├── Gallery/
│   └─ Home/
└─ downloads/
    ├── BrightWorld-Price-List.html
    ├── BrightWorld-Product-Catalogue.html
    ├── BrightWorld-Technical-Specs.html
    └─ BrightWorld-Warranty-Guide.html
```

Design System & Features

Color Palette

Variable	Color Code	Usage
--primary	#0a84ff	Buttons, links, active states, glow effects
--accent	#f5c518	Highlights, badges, star ratings, CTA text
--dark	#0d0f1a	Dark mode page background
--dark-card	#1a1d30	Card and component backgrounds
--text-main	#f0f4ff	Primary text in dark mode
--glass-bg	rgba(255,255,255,0.06)	Glassmorphism card backgrounds

Typography

Font Family	Weights	Applied To
Inter	300–900	Body text, paragraphs, UI labels, prices
Poppins	300–900	Section headings, page titles
Montserrat	400–900	Brand name, logo text, hero headlines

Key Features Implemented

- **Dark / Light Mode Toggle** — Theme preference saved in browser LocalStorage across sessions.
- **Loading Screen** — Animated BrightWorld logo loader shown on every page (1.8 seconds).
- **Sticky Navbar** — Navbar compacts with backdrop blur effect after 60px scroll.
- **Hero Carousel** — 4 auto-rotating full-screen slides with gradient LED themes.
- **Animated Counters** — Stats (10K+ Customers etc.) count up when scrolled into viewport.
- **Real-time Product Search** — Results filter instantly as user types in the search box.
- **Category Filter** — Single-click filter buttons to show/hide products by category.
- **Product Sort** — Sort by Price (Low→High / High→Low) or Name (A→Z / Z→A).
- **Shopping Cart** — Add/remove items, quantity management, LocalStorage

persistence.

- **URL Deep Linking** — Direct URL filter support: `products.html?filter=bulbs`
- **AOS Animations** — Scroll-triggered animations (700ms, ease-out-cubic).
- **Mobile Responsive** — Bootstrap 5 grid + offcanvas menu for screens below 992px.
- **Scroll-to-Top Button** — Floating button appears after 300px scroll.

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Source Code

HTML — Navigation Bar (index.html)

index.html — Navbar

```
<nav class="navbar navbar-expand-lg sticky-top" id="mainNav">
  <div class="container">
    <a class="navbar-brand d-flex align-items-center" href="index.html">
      <div class="logo-icon">
        <i class="bi bi-lightbulb-fill text-white"></i>
      </div>
      Bright<span class="brand-world">World</span>
    </a>
    <div class="collapse navbar-collapse" id="navbarNav">
      <ul class="navbar-nav mx-auto gap-1">
        <li class="nav-item"><a class="nav-link" href="index.html">Home</a></li>
        <li class="nav-item dropdown">
          <a class="nav-link dropdown-toggle" href="#" data-bs-toggle="dropdown">
            Shop
          </a>
          <ul class="dropdown-menu">
            <li><a class="dropdown-item" href="products.html">Products</a></li>
            <li><a class="dropdown-item" href="brands.html">Brands</a></li>
            <li><a class="dropdown-item" href="gallery.html">Gallery</a></li>
            <li><a class="dropdown-item" href="download.html">Downloads</a></li>
          </ul>
        </li>
      </ul>
      <a href="cart.html" class="btn-primary-grad btn btn-sm px-3 py-2">
        <i class="bi bi-cart3"></i> Cart
        <span class="badge bg-warning" id="cart-count">0</span>
      </a>
    </div>
  </div>
</nav>
```

CSS — CSS Custom Properties (style.css)

css/style.css — Root Variables

```
:root {
  --primary:      #0a84ff;
  --primary-dark: #005ecb;
  --accent:       #f5c518;
  --dark:         #0d0f1a;
  --dark-card:    #1a1d30;
  --surface:      #1e2238;
  --text-main:    #f0f4ff;
  --text-muted:   #8892b0;
  --glass-bg:     rgba(255,255,255,0.06);
  --glass-border: rgba(255,255,255,0.12);
  --shadow-blue:  0 8px 32px rgba(10,132,255,0.25);
  --radius-md:    14px;
  --transition:   all 0.35s cubic-bezier(0.4,0,0.2,1);
  --font-main:    'Inter', sans-serif;
  --font-heading: 'Poppins', sans-serif;
  --font-brand:   'Montserrat', sans-serif;
}

[data-theme="light"] {
  --dark:      #f0f4ff;
  --dark-card: #ffffff;
  --text-main: #0d0f1a;
  --text-muted: #4a5568;
  --nav-bg:    rgba(240,244,255,0.97);
}
```

Source Code (Continued)

JavaScript — Dark/Light Mode Toggle (script.js)

js/script.js — Theme Toggle

```
/* Dark / Light Mode Toggle */
const darkToggleBtn = document.getElementById('dark-toggle');
const currentTheme = localStorage.getItem('bw-theme') || 'dark';
document.documentElement.setAttribute('data-theme', currentTheme);
updateToggleIcon(currentTheme);

if (darkToggleBtn) {
  darkToggleBtn.addEventListener('click', () => {
    const theme = document.documentElement
      .getAttribute('data-theme') === 'dark' ? 'light' : 'dark';
    document.documentElement.setAttribute('data-theme', theme);
    localStorage.setItem('bw-theme', theme);
    updateToggleIcon(theme);
  });
}

function updateToggleIcon(theme) {
  if (!darkToggleBtn) return;
  darkToggleBtn.innerHTML = theme === 'dark'
    ? '<i class="bi bi-sun-fill"></i>'
    : '<i class="bi bi-moon-fill"></i>';
}
```

JavaScript — Product Filter System (script.js)

js/script.js — Product Filter

```

/* Product Category Filter */
const filterBtns = document.querySelectorAll('.filter-btn');
filterBtns.forEach(btn => {
  btn.addEventListener('click', () => {
    filterBtns.forEach(b => b.classList.remove('active'));
    btn.classList.add('active');
    const filter = btn.dataset.filter;
    const items = document.querySelectorAll('.filter-item');
    items.forEach(item => {
      const show = filter === 'all' || item.dataset.category === filter;
      item.style.display = show ? '' : 'none';
      if (show) { item.style.animation = 'fadeIn 0.4s ease'; }
    });
  });
});

/* Real-time Product Search */
const productSearch = document.getElementById('product-search');
if (productSearch) {
  productSearch.addEventListener('input', () => {
    const q = productSearch.value.toLowerCase().trim();
    document.querySelectorAll('.filter-item').forEach(item => {
      const name = (item.querySelector('.product-name')
        ? item.textContent || '' : '').toLowerCase();
      item.style.display = name.includes(q) ? '' : 'none';
    });
  });
}

```

JavaScript — Shopping Cart (script.js)

js/script.js — Cart System

```

/* Cart System */
let cart = JSON.parse(localStorage.getItem('bw-cart') || '[]');

function updateCartCount() {
  const badge = document.getElementById('cart-count');
  if (badge) badge.textContent = cart.reduce((s, i) => s + i.qty, 0);
}
updateCartCount();

function addToCart(name, price, brand) {
  const existing = cart.find(i => i.name === name);
  if (existing) {
    existing.qty += 1;
  } else {
    cart.push({ name, price, brand, qty: 1 });
  }
  localStorage.setItem('bw-cart', JSON.stringify(cart));
  updateCartCount();
}

```

Testing & Results

The BrightWorld website has been thoroughly tested across multiple browsers, screen sizes, and functional scenarios. The following table summarizes the testing results:

#	Test Case	Expected Result	Result
1	All 11 pages load without errors	Pages load correctly in browser	✓ Pass
2	Navbar dropdown menus open/close	Dropdowns show on click	✓ Pass
3	Mobile offcanvas menu works	Side menu opens on small screens	✓ Pass
4	Dark / Light mode toggle	Theme switches correctly	✓ Pass
5	Theme saved after page refresh	Same theme on reload	✓ Pass
6	Hero carousel auto-rotates	Slides change every 5 seconds	✓ Pass
7	Product category filter buttons	Shows only selected category	✓ Pass
8	Real-time product search	Filters results as user types	✓ Pass
9	Product sort (price / name)	Products reorder correctly	✓ Pass
10	Add to Cart button	Item added to cart	✓ Pass
11	Cart count badge updates	Badge shows correct count	✓ Pass
12	Cart persists after refresh	Items still in cart after reload	✓ Pass
13	Feedback form renders	Form fields display correctly	✓ Pass
14	Download links work	Documents open on click	✓ Pass
15	Gallery images display	Images load with hover effects	✓ Pass
16	AOS animations on scroll	Elements animate in smoothly	✓ Pass
17	Animated stats counters	Numbers count up on scroll	✓ Pass

#	Test Case	Expected Result	Result
18	Responsive on mobile (375px)	Layout adapts correctly	✓ Pass
19	Responsive on tablet (768px)	Layout adapts correctly	✓ Pass
20	URL parameter filter works	products.html?filter=bulbs works	✓ Pass
21	Brand URL filter works	products.html?brand=Philips works	✓ Pass
22	Footer navigation links	All footer links navigate correctly	✓ Pass
23	Contact page information	Phone, email, hours correct	✓ Pass
24	Scroll-to-top button	Button appears, scrolls to top	✓ Pass
25	Cross-browser compatibility	Works in Chrome, Edge, Firefox	✓ Pass

Testing Summary: 25 out of 25 test cases passed. No critical bugs found.

Conclusion

The **BrightWorld LED Lights Website** has been successfully designed, developed, tested, and documented as a complete E-Project submission for Aptech Computer Education's HDSE Semester III program. This project represents a comprehensive, fully functional, and professionally crafted web-based solution that fulfills all 10 client requirements as specified in the project brief.

The website serves as a complete digital storefront for the BrightWorld LED store, providing customers with a seamless online experience to browse over multiple product categories, explore top LED brands, view specifications and pricing, access store information, submit feedback, and download product documentation — all through an elegantly designed and fully responsive web interface.

From a technical standpoint, the project demonstrates strong proficiency in modern web development technologies including **HTML5, CSS3, JavaScript (ES6+), Bootstrap 5, AOS Animations, Google Fonts**, and browser-based LocalStorage API. The implementation of features such as real-time search, dynamic product filtering, animated counters, a persistent shopping cart, and a Dark/Light mode toggle reflects a high standard of practical web development skill.

The project also highlights the importance of user experience design — with glassmorphism effects, smooth animations, hover interactions, and a premium dark-mode aesthetic that creates a visually impressive and engaging digital presence for the BrightWorld brand.

Through this project, our team gained invaluable hands-on experience in:

- Translating client requirements into a fully functional and deployable web application.
- Implementing responsive design principles for cross-device compatibility.
- Applying JavaScript for dynamic UI interactions, state management, and data persistence.
- Working collaboratively as a team with clear task delegation and coordination.
- Writing professional project documentation following academic and industry standards.

In conclusion, the BrightWorld E-Project not only meets all defined requirements but goes beyond the minimum specifications to deliver a premium-quality product. It bridges the gap between academic learning and professional web development practice, and we are confident it stands as a strong representation of our skills and dedication.

We look forward to the evaluation and feedback from the E-Projects Team at Aptech and remain committed to learning, growing, and applying our knowledge in future endeavors.

— Thank You —

MR. HUSSNAIN AHMED | MR. MUHAMMAD ALI | MR. AHMED RAZA | MR.
USMAN TARIQ

HDSE Semester III — E-Project | Aptech Computer Education | July 2026

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